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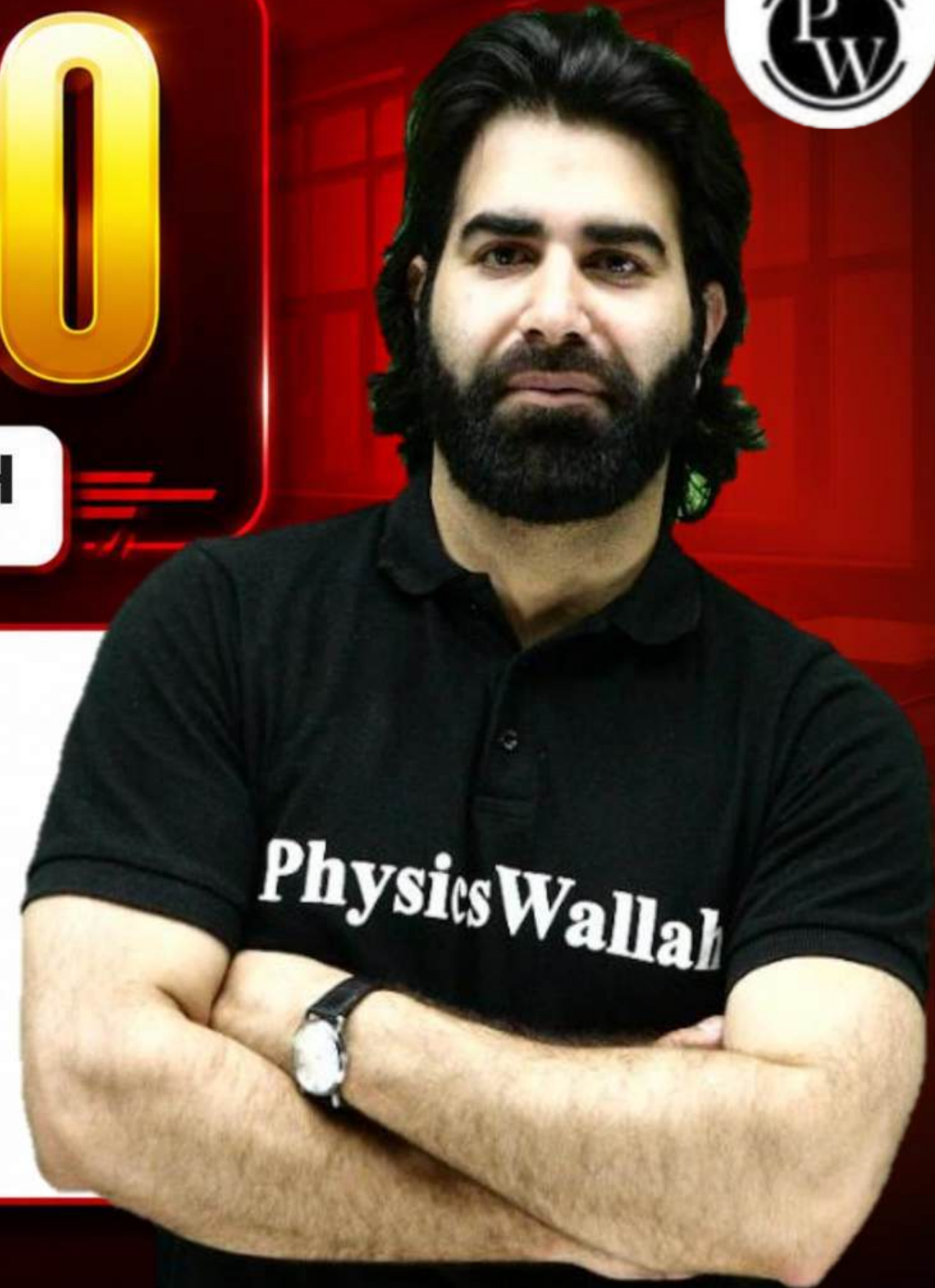
JKBOSE CLASS-11TH

Breathing and Gaseous Exchange

One Shot

Zoology.

Adil Altaf Sir



Topics To Be Covered



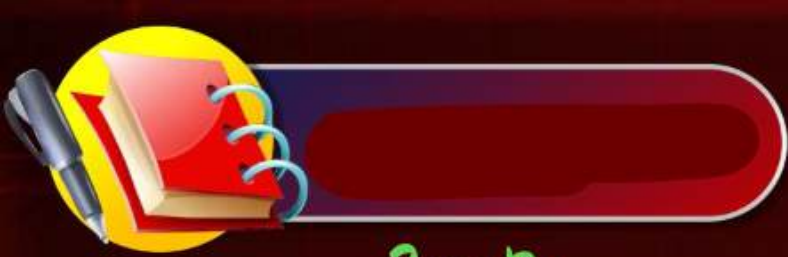
Human Physiology
→ Gross Functioning.

1 Breathing and Gaseous Exchange

2 - Human Respiratory Imp. ()

3 - Mechanism of Breathing (✓)
- Lung Volumes & Capacities

4 - Transport of Gases
- Regulation Disorders



Breathing

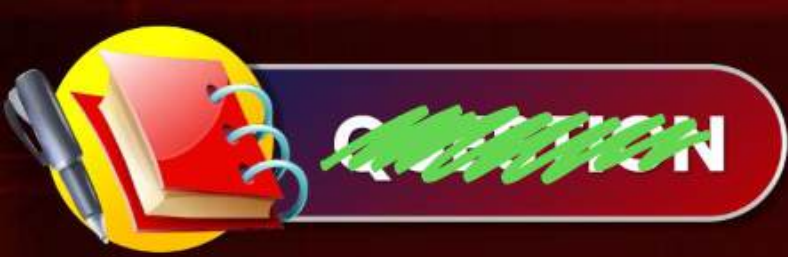
The process of exchange of O_2 from the atmosphere with CO_2 produced by the cells is called Breathing.
ncert - This is Also klas (Respiration) [physical Respiration]

Respiration - A process in which oxidation of organic compounds occurs in cell and energy is released is known as Respiration.
(Food)

Types Of Respiration:

1. Branchial Respiration - It is the type of respiration which occurs through gills. For e.g. - Fishes.
2. Cutaneous Respiration - It is a form of respiration in which occurs across the skin or outer integument of an organism. For e.g. - Frogs.
3. Pulmonary Respiration - It is the type of respiration which occurs through lungs. For e.g. - Humans.

Respiratory Organs | Organisms



①

Respiratory Organs	Organisms
<u>Entire Body surface</u>	Sponges, ✓ flatworms. ✓ ✓ coelenterate,
<u>Skin</u> (cutaneous)	<u>Earthworm.</u>
<u>Tracheal system</u>	<u>Insects</u>
<u>Gills</u>	<u>Pisces</u> , aquatic arthropods.
<u>Lungs</u>	<u>Amphibians</u> , <u>mammals</u> . Reptiles, Birds



Human Respiratory System [Nose, Pharynx, Larynx, Trachea, Bronchi, Bronchioles, Alveoli]

Nose

1. Nostrils – A pair of external nostrils are present above the upper lip.
2. Nasal Chamber – Nostrils leads to a nasal chamber through the nasal passage. It opens into the pharynx.

3. Pharynx – It is the common passage for food and air.

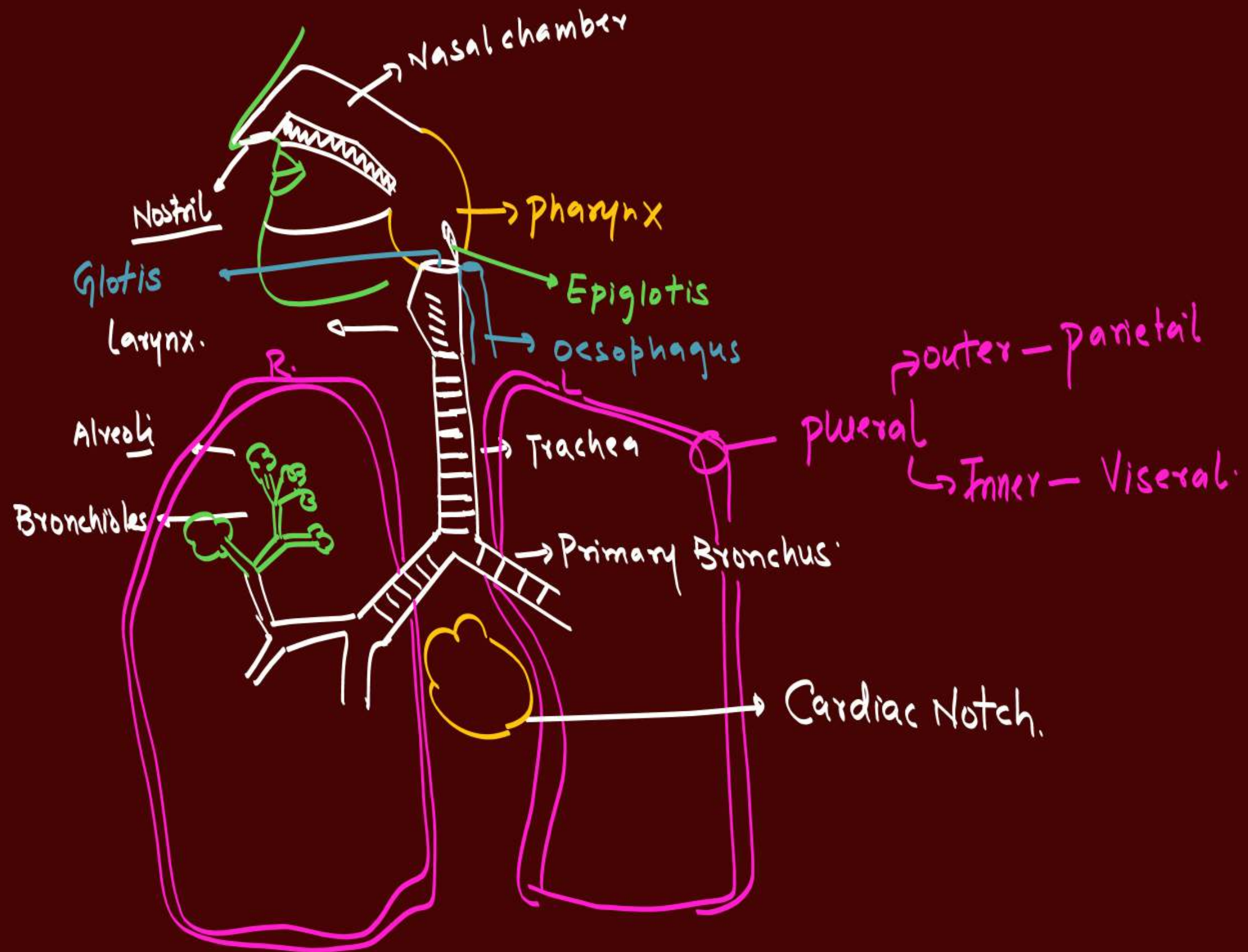
[It Receives Mouth (oral cavity) and Nasal cavity]

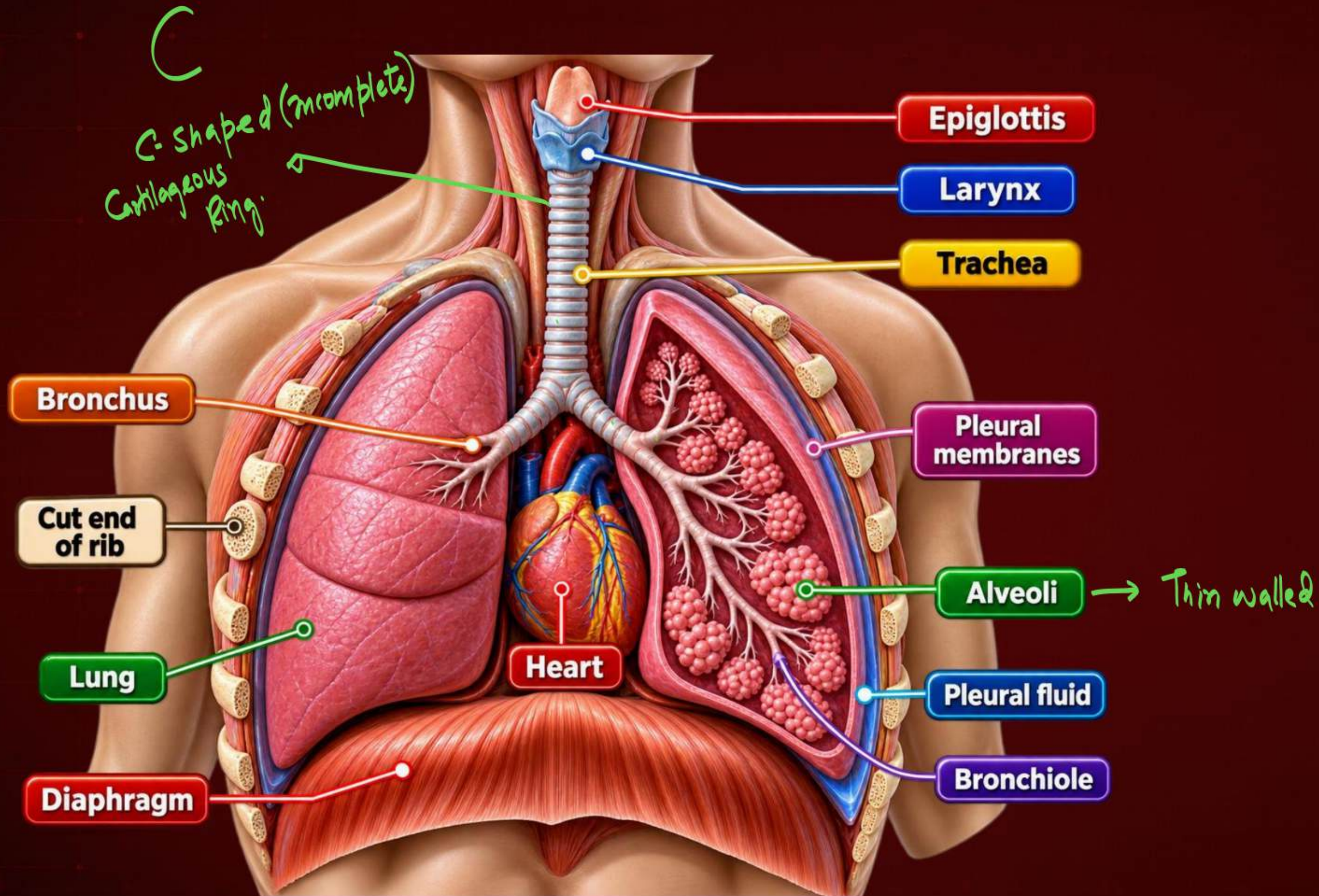
4. Glottis – A part of larynx.

Larynx

5. Larynx – The pharynx opens through the larynx region into the trachea.

Larynx is a cartilaginous box which helps in sound production and hence called the Sound Box.





C-shaped (incomplete)
Cartilaginous Ring

Squamous Epithelium

Thin walled



6. Epiglottis – During swallowing glottis can be covered by a thin elastic cartilaginous flap called epiglottis to prevent the entry of food into the larynx.
7. Trachea – Trachea is a straight tube extending up to the mid-thoracic cavity, which divides at the level of 5th thoracic vertebra into a right and left primary bronchi.
8. Bronchi – Each bronchi undergoes repeated divisions to form the secondary and tertiary bronchi and bronchioles ending up in very thin terminal bronchioles. The tracheae, primary, secondary and tertiary bronchi, and initial bronchioles are supported by incomplete cartilaginous rings.



9. Alveoli – Each terminal bronchiole gives rise to a number of very thin, irregular-walled and vascularized bag-like structures called alveoli. (Blood supply)
10. Pleural membrane – lungs are covered by a double layered pleura, with pleural fluid between them. It reduces friction on the lung-surface.
- The outer pleural membrane is in close contact with the thoracic lining whereas the inner pleural membrane is in contact with the lung surface.

(Parts of Respiratory system)

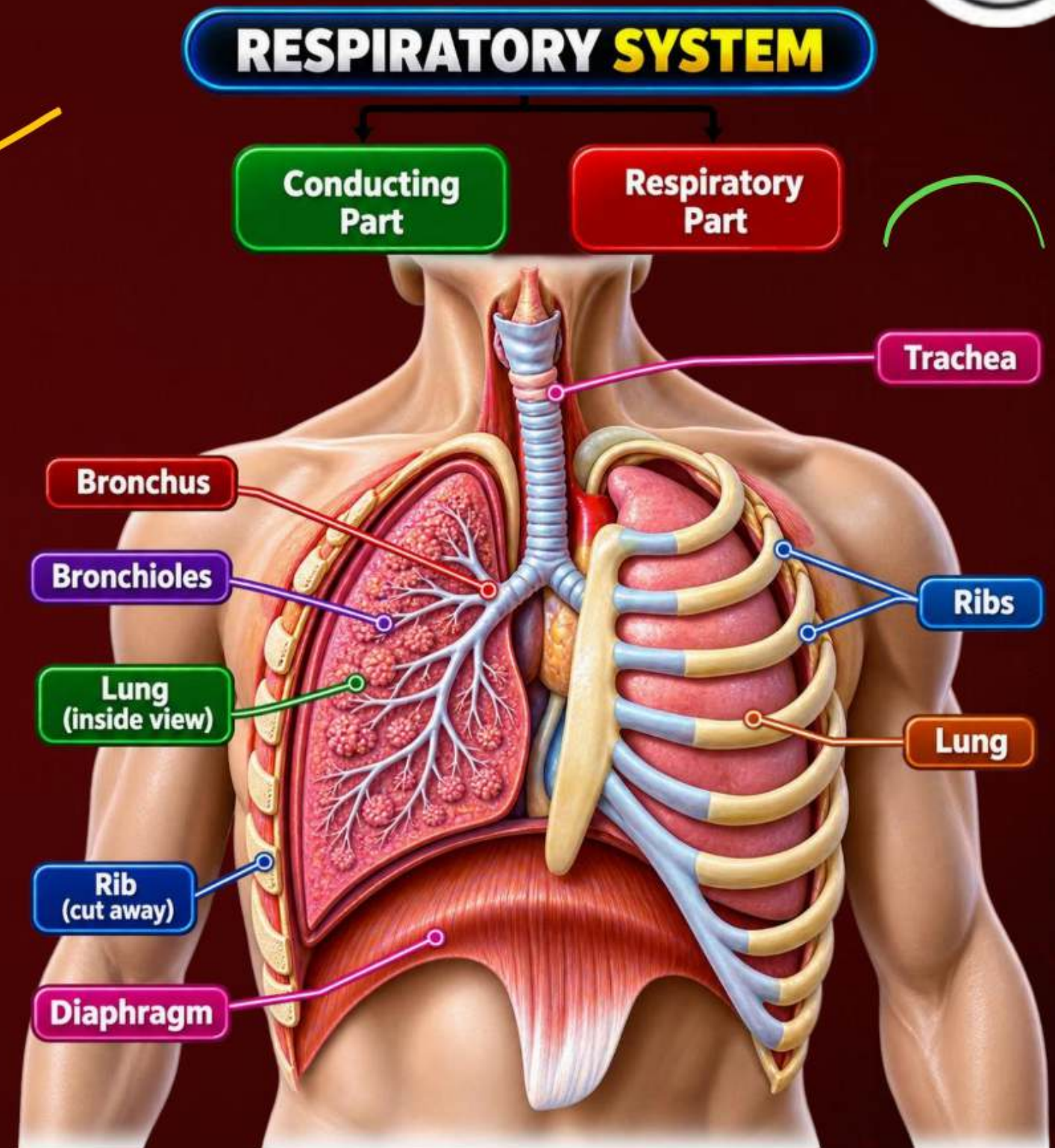
1. Conducting Part: The part starting with the external nostrils up to the terminal bronchioles constitute the conducting part.



- It transports the atmospheric air to the alveoli, clears it from foreign particles, humidifies and also brings the air to body temperature.

2. **Respiratory Part:** The alveoli and their ducts form the respiratory or exchange part of the respiratory system.

- It is the site of actual diffusion of O_2 and CO_2 between blood and atmospheric air.





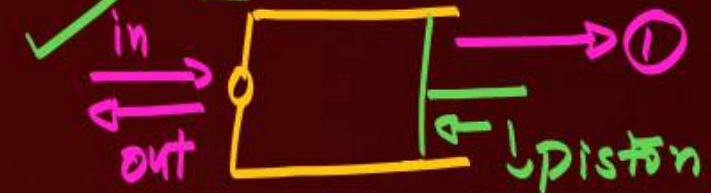
Topic - Mechanism of Breathing

Boyle's Law
$$\left[\underset{\downarrow}{\text{Pressure}} \times \underset{\uparrow}{\text{Volume}} = \text{Const} \right] \rightarrow \begin{matrix} \text{Change in Vol} \\ \text{change in pressure} \end{matrix}$$

①
②



- The Lungs are situated in the thoracic chamber which is anatomically an air-tight chamber.
- The thoracic chamber is formed dorsally by the vertebral column, ventrally by the sternum, laterally by the ribs and on the lower side by the dome-shaped diaphragm.
- The anatomical setup of lungs in thorax is such that any change in the volume of the thoracic cavity will be reflected in the lung (pulmonary) cavity.
- Such an arrangement is essential for breathing, as we cannot directly alter the pulmonary volume.





Mechanism Of Respiration: -

Breathing or pulmonary ventilation by which atmospheric air is drawn in and CO₂ rich alveolar air is released out.

Diffusion of gases (O₂ and CO₂) across alveolar membrane.

Transport of gases by the blood.

Diffusion of O₂ and CO₂ between blood and tissues.

Utilisation of O₂ by the cells for catabolic reactions and resultant release of CO₂.

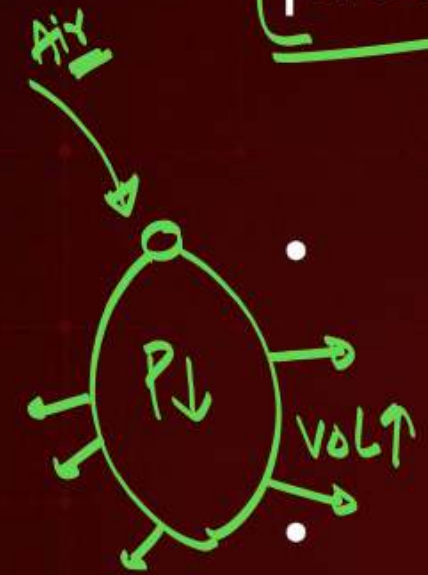
Mechanism Of Breathing: -

Breathing involves two stages: Inspiration during which atmospheric air is drawn in and Expiration by which the alveolar air is released out. The



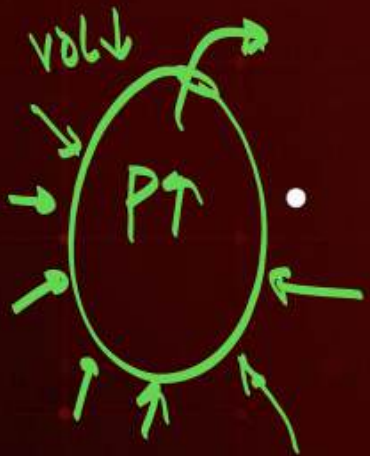
QUESTION

Movement of air into and out of the lungs is carried out by creating a pressure gradient between the lungs and atmosphere.



- Inspiration can occur if the pressure within the lungs (intra-pulmonary pressure) is less than the atmospheric pressure.

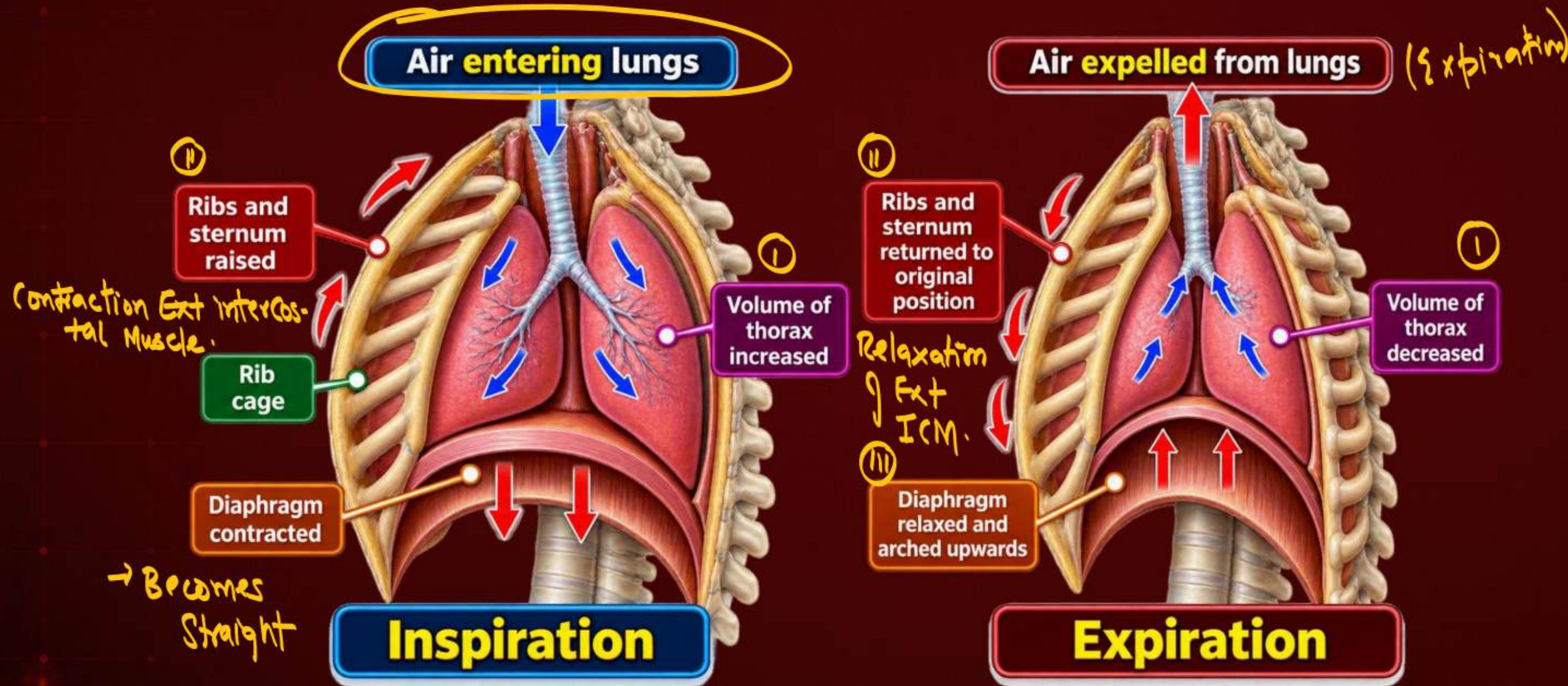
Expiration takes place when the intra-pulmonary pressure is higher than the atmospheric pressure.



- The Diaphragm and a specialised set of muscles – external and internal intercostals between the ribs, help in generation of such gradients.

- On an average, a healthy human breathes 12–16 times/minute.

The volume of air involved in breathing movements can be estimated by using a Spirometer which helps in clinical assessment of pulmonary functions.



Trick

2) Muscles

- Diaphragm
- Ext Intercostal

ICER → Relaxation

Contraction



Respiratory Volumes And Capacities

→ Lung Functional Test (Spirometry)

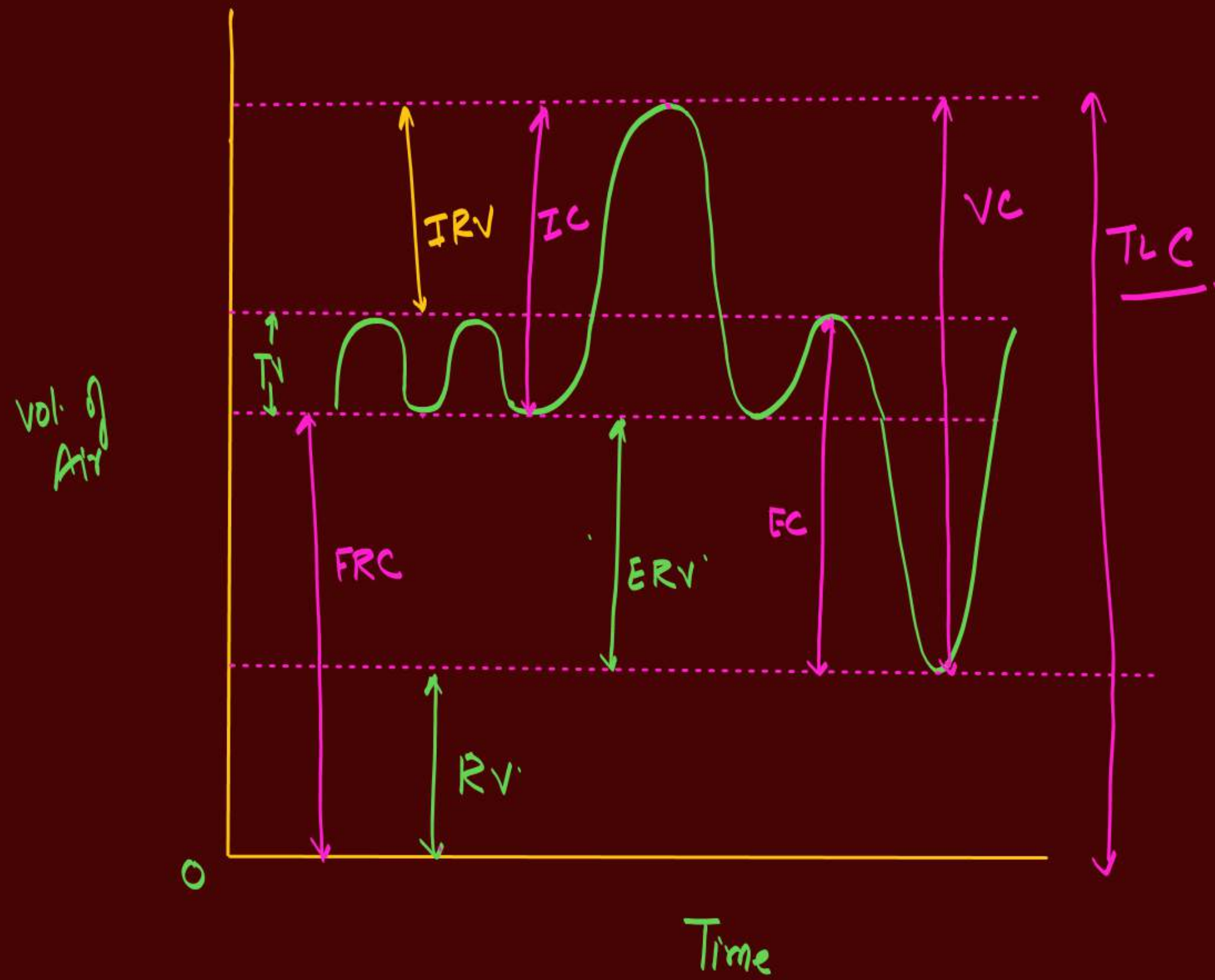
Device → Measure - Person's Amount of Air

Lung Volumes (V)

1. Tidal Volume: Volume of air inspired or expired during a normal respiration. It is approx. 500 mL, i.e., a healthy man can inspire or expire approximately 6000 to 8000 mL of air per minute.
2. Inspiratory Reserve Volume: Additional volume of air, a person can inspire by a forcible inspiration. This averages 2500 mL to 3000 mL.
3. Expiratory Reserve Volume: Additional volume of air, a person can expire by a forcible expiration. This averages 1000 mL to 1100 mL.
4. Residual Volume: Volume of air remaining in the lungs even after a forcible expiration. This averages 1100 mL to 1200 mL.
5. Inspiratory Capacity: Total volume of air a person can inspire after a normal expiration. This includes tidal volume and inspiratory reserve volume (TV + IRV).

12-16 times/Min

Combining volumes,
gives us
Lung Capacities





Respiratory Volumes And Capacities

6. **Expiratory Capacity:** Total volume of air a person can expire after a normal inspiration. This includes tidal volume and expiratory reserve volume ($TV + ERV$).
7. **Functional Residual Capacity:** Volume of air that will remain in the lungs after a normal expiration. This includes ($ERV + RV$).
8. **Vital Capacity:** The maximum volume of air a person can breathe in after a forced expiration. This includes ERV, TV and IRV or the maximum volume of air a person can breathe out after a forced inspiration.
9. **Total Lung Capacity:** Total volume of air accommodated in the lungs at the end of a forced inspiration. This includes RV, ERV, TV and IRV or vital capacity + residual volume.



Partial Pressure Values

Q How do gases get exchanged - At tissues
- At lungs

Answer - Due to Difference in their Partial Pressures (from Higher P.P to lower).



The partial pressure of O_2 in alveolar air is 104 mm Hg and its value in blood is 40 mm Hg. So, oxygen goes from alveoli to blood. Now, partial pressure of O_2 in blood is 95 mm Hg and its value in tissue is 40 mm Hg. So, oxygen goes from blood to tissues.

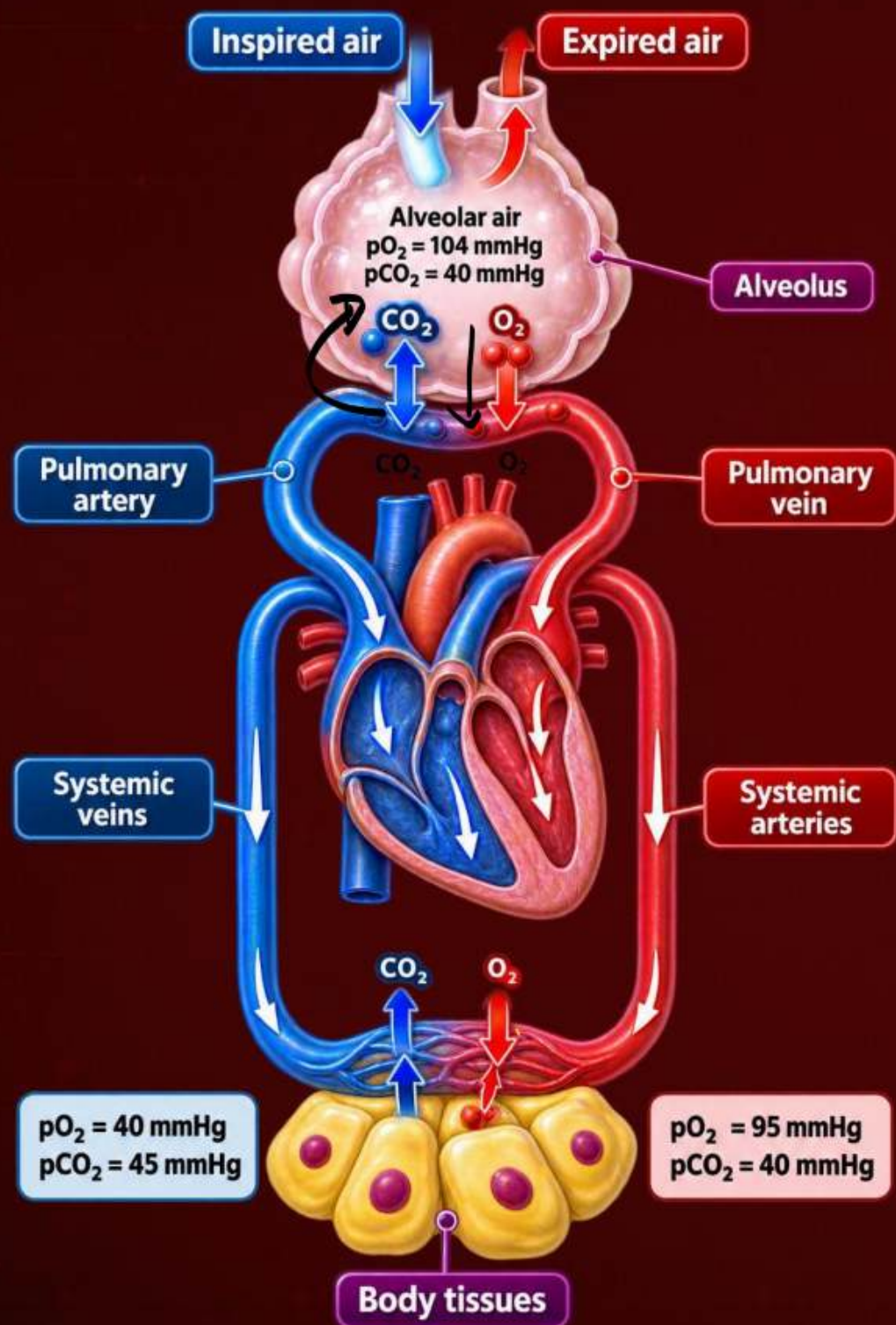
(In mm Hg)

Respiratory gas	Atmospheric air	Alveoli	Blood deoxygenated	Blood oxygenated	Tissues
O_2	159	104	40	95	40
CO_2	0.3	40	45	40	45

Partial P
"



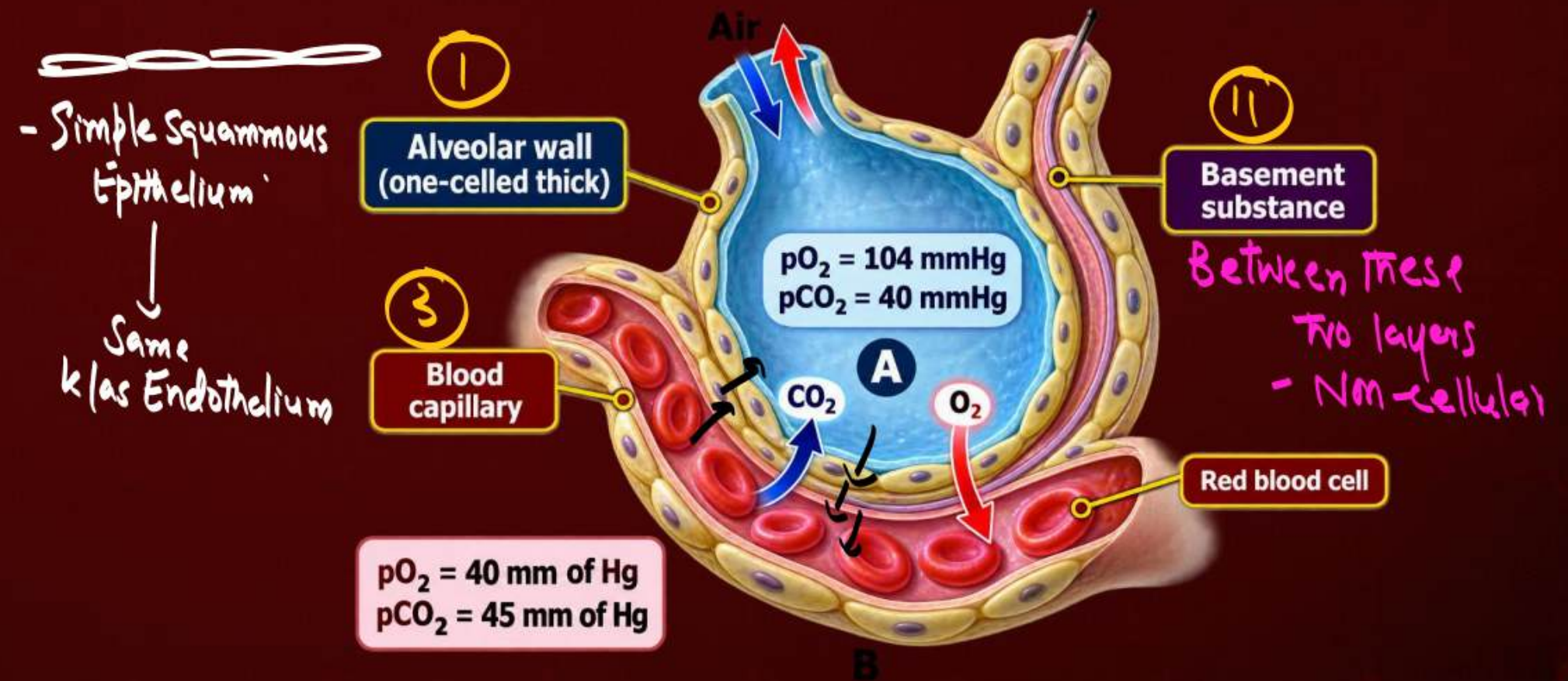
Partial Pressure Values



QUESTION

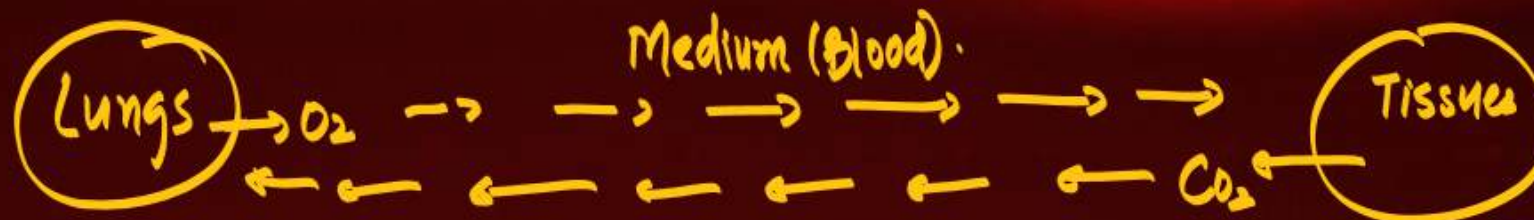


The partial pressure of CO_2 in alveolar air is 40 mm Hg and its value in blood is 45 mm Hg. So, oxygen goes from blood to alveoli. Now, partial pressure of CO_2 in blood is 45 mm Hg and its value in oxygenated blood is 40 mm Hg. So, oxygen goes from tissues to blood.





Transport Of Gases



- Oxygen • Blood is the medium of transport for O_2 and CO_2 . About 97% of O_2 is transported by RBCs in the blood.
- The remaining 3% of O_2 is carried in a dissolved state through the Plasma. (liquid content of blood mainly H_2O)
- CO_2 • Nearly 20–25% of CO_2 is transported by RBCs whereas 70% of it is carried as Bicarbonate.
- About 7% of CO_2 is carried in a dissolved state through plasma.

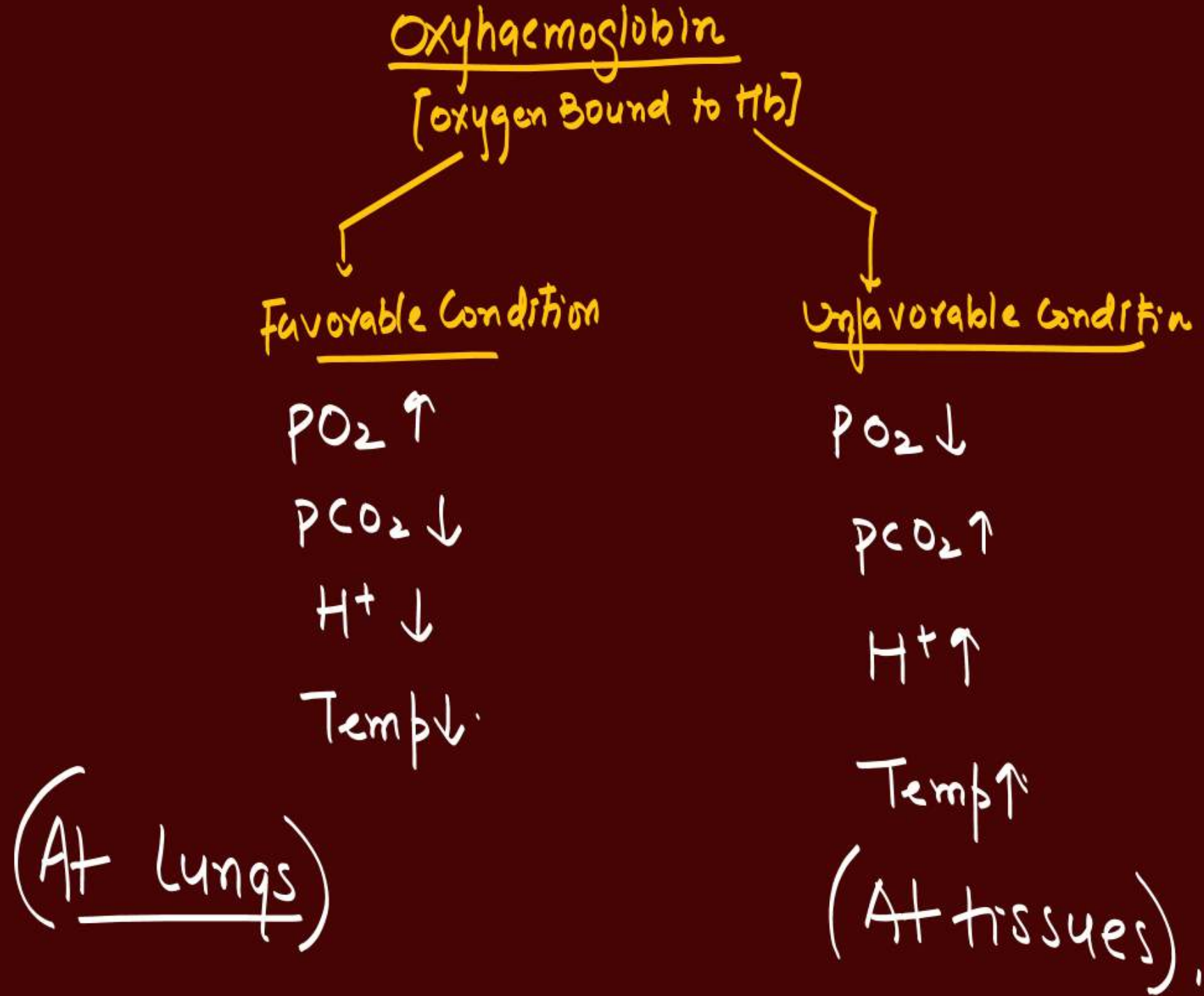
Transport Of Oxygen

- Haemoglobin is a red coloured iron containing pigment present in the RBCs.



Transport Of Oxygen

- ii. O₂ can bind with haemoglobin in a reversible manner to form Oxyhaemoglobin.
- iii. Each haemoglobin molecule can carry a maximum of four molecules of O₂. Binding of oxygen with haemoglobin is primarily related to partial pressure of O₂.
- iv. Partial pressure of CO₂, hydrogen ion concentration and temperature are the other factors which can interfere with this binding.
- v. A Sigmoid Curve is obtained when percentage saturation of haemoglobin with O₂ is plotted against the pO₂. This curve is called the Oxygen Dissociation Curve.



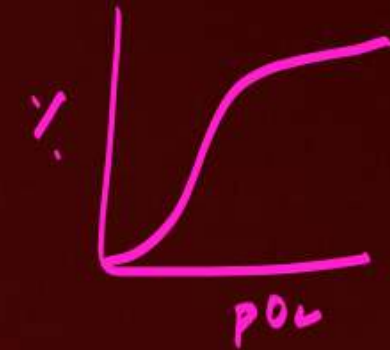
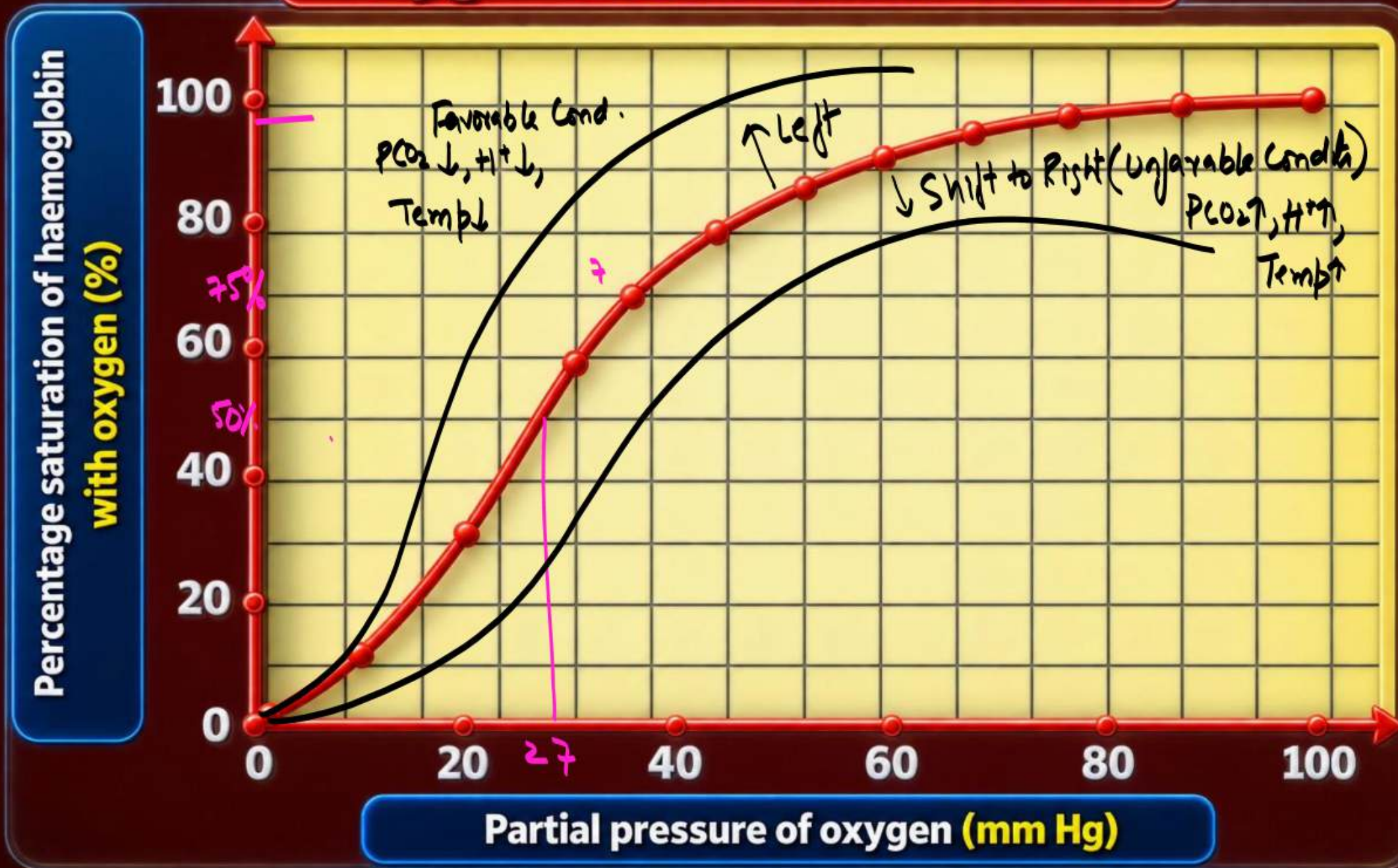


Transport Of Oxygen



Oxygen Dissociation Curve

→ S-shaped (Sigmoid curve)





Transport Of Oxygen



vi. In the Alveoli, where there is high pO_2 , low pCO_2 , lesser H^+ concentration and low temperature, the factors are all favourable for the formation of oxyhaemoglobin, whereas in the Tissues, where low pO_2 , high pCO_2 , high H^+ concentration and high temperature exist, the conditions are favourable for dissociation of oxygen from the oxyhaemoglobin.

vii. This clearly indicates that O_2 gets bound to haemoglobin in the lung surface and gets dissociated at the tissues. Every 100 ml of Oxygenated Blood can deliver around 5 ml of O_2 to the tissues under normal physiological conditions.



Transport of Carbon Dioxide

→ 3 ways →
→ 20-25%
→ 70%
→ 7%

CO₂ is Attached to Amino groups of protein chain

i. CO₂ is carried by haemoglobin as Carbamino-Haemoglobin (about 20-25%). This binding is related to the partial pressure of CO₂.

High (Tissues) Low (Alveoli)

ii. When pCO₂ is high and pO₂ is low as in the Tissues, more binding of CO₂ occurs whereas, when the pCO₂ is low and pO₂ is high as in the Alveoli, dissociation of CO₂ from carbamino-haemoglobin takes place, i.e., CO₂ which is bound to haemoglobin from the tissues is delivered at the alveoli.

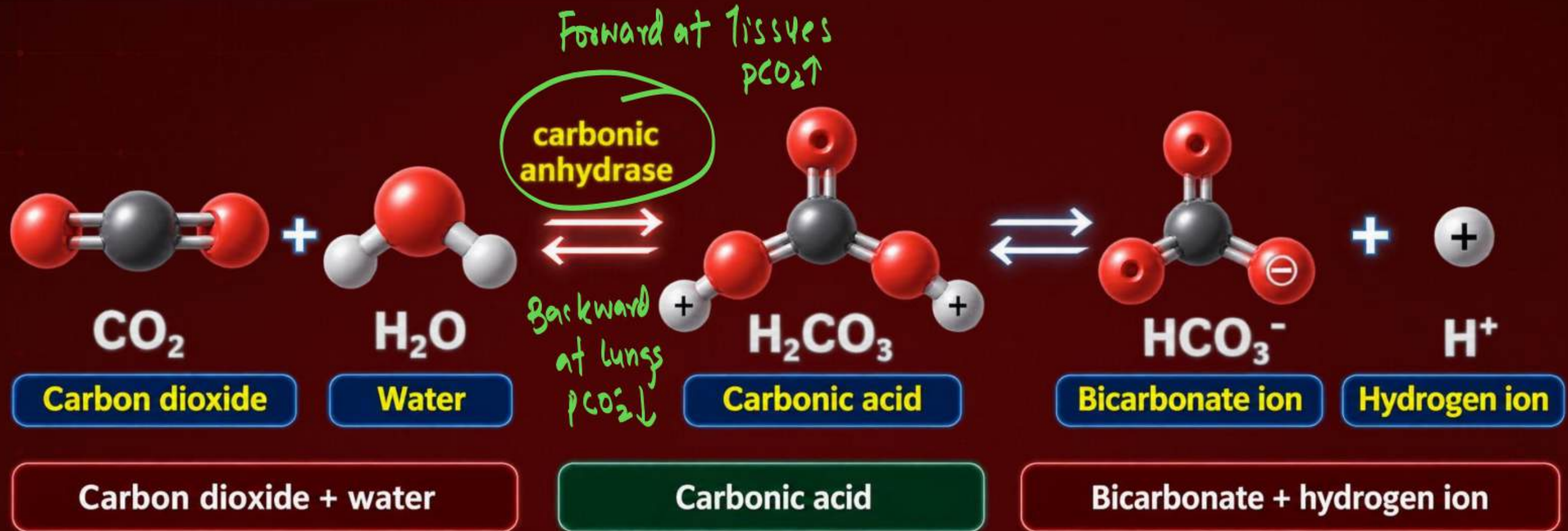
– Majority of CO₂ is transported as Bicarbonate (HCO₃⁻)



iii. RBCs contain a very high concentration of the enzyme, Carbonic Anhydrase and minute quantities of the same is present in the plasma too. This enzyme facilitates the following reaction in both directions.



Transport of Carbon Dioxide





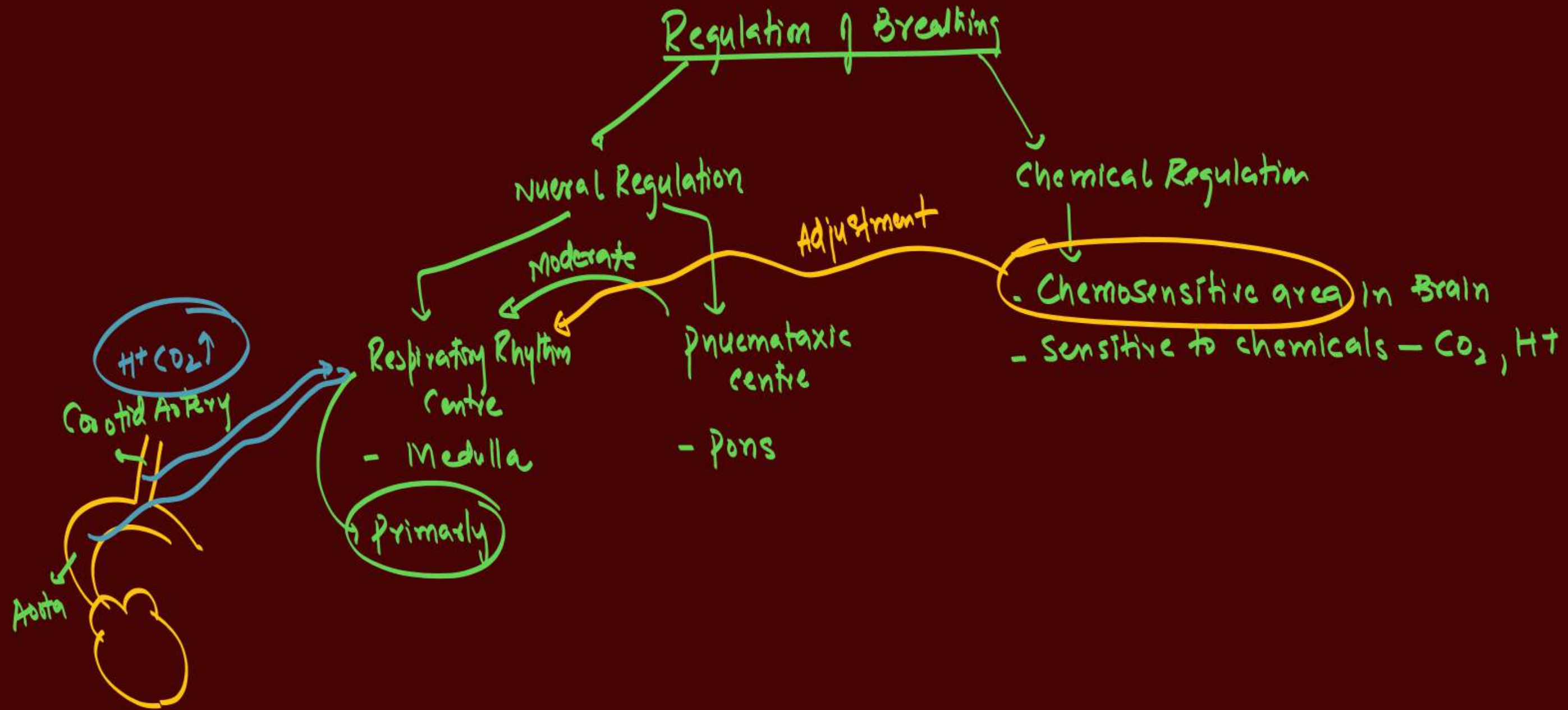
Transport of Carbon Dioxide

- iv. At the tissue site where partial pressure of CO_2 is high due to catabolism, CO_2 diffuses into blood (RBCs and Plasma) and forms HCO_3^- and H^+ .
- v. At the alveolar site where pCO_2 is low, the reaction proceeds in the opposite direction leading to the formation of CO_2 and H_2O .
- vi. Thus, CO_2 trapped as bicarbonate at the tissue level and transported to the alveoli is released out as CO_2 . Every 100 ml of deoxygenated blood delivers approximately 4 ml of CO_2 to the alveoli.



Regulation Of Respiration

- Human beings have a significant ability to maintain and moderate the Respiratory Rhythm to suit the demands of the body tissues. This is done by the Neural System.
- A specialised centre present in the Medulla of the brain called Respiratory Rhythm Centre is primarily responsible for this regulation.
- Another centre present in the Pons of the brain called Pneumotaxic Centre can moderate the functions of the respiratory rhythm centre.
- Neural signal from this centre can reduce the duration of inspiration and thereby alter the respiratory rate.





Regulation Of Respiration



- A Chemosensitive Area is situated adjacent to the rhythm centre, which is highly sensitive to CO_2 and hydrogen ions. H^+
- Increase in these substances can activate this centre, which in turn can signal the rhythm centre to make necessary adjustments in the respiratory process.
- Receptors associated with Aortic Arch and Carotid Artery also can recognise changes in CO_2 and H^+ concentration and send necessary signals to the rhythm centre for remedial actions.



Disorders Of Respiratory System



- Asthma: is a difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles. *Allergic Disorder*
- Emphysema: is a chronic disorder in which alveolar walls are damaged due to which ^{Slowly time taken} respiratory surface is decreased. One of the major causes of this is cigarette smoking.
- Occupational Respiratory Disorders: In certain industries, especially those involving grinding or stone-breaking, so much dust is produced that the defense mechanism of the body cannot fully cope with the situation. Long exposure can give rise to inflammation leading to Fibrosis (proliferation of fibrous tissues) and thus causing serious lung damage. Workers in such industries should wear protective masks.



Thank
You